AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 10 · WORKBOOK

Evaluation pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete evaluation-pipeline evidence record.

Keep planned IDs, explicit statuses, references, actual outputs, versions and declared rates together.

Keep one habit: Recall → W1–W3/W5 → independent rate calculation → project evaluation manifest → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Evaluation pipelines preserve case versions, procedure versions and **raw outcome**s before producing metrics. A rerun must identify what changed.

- Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Integration tests and fixtures

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Testing pure components

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Configuration and secrets

> **My remembered idea, example and correction**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main manifest: six planned room-capacity cases C1–C6; case version classroom-v1, fixed lookup stub-v1. Inputs A/B/C/D/E/F have reference 12/8/10/6/4/2. Completed actual outputs 12/8/10/6/5 for C1–C5 yield four passes and one fail; C6 explicitly skipped with reason, not run. Completed pass rate 4/5=80%, coverage 5/6≈83.33%; skip remains unassessed.

  One explicit record per planned ID required; missing, duplicate, unknown IDs or unexpected fields reject the record set. Wrong actual output type/value is a completed failure. All explicit skips give coverage 0 and undefined completed pass rate. A stub replay rechecks the same procedure, not independent data.

  Separate picture correctness flags A[1, 1, 0, 1], B[1, 0, 0, 1] yield 75%/50% on four shared cases; case 2 differs, case 3 failsboth. Deleting case 3 post-scoring yields 3/3 and2/3 but conceals failure if replacing original. Keep versions, raw outcomes and declared subgroup rules. No live model or broader quality measurement supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main manifest: six planned room-capacity cases C1–C6; case version classroom-v1, fixed lookup stub-v1. Inputs A/B/C/D/E/F have reference 12/8/10/6/4/2. Completed actual outputs 12/8/10/6/5 for C1–C5 yield four passes and one fail; C6 explicitly skipped with reason, not run. Completed pass rate 4/5=80%, coverage 5/6≈83.33%; skip remains unassessed.

  One explicit record per planned ID required; missing, duplicate, unknown IDs or unexpected fields reject the record set. Wrong actual output type/value is a completed failure. All explicit skips give coverage 0 and undefined completed pass rate. A stub replay rechecks the same procedure, not independent data.

  Separate picture correctness flags A[1, 1, 0, 1], B[1, 0, 0, 1] yield 75%/50% on four shared cases; case 2 differs, case 3 failsboth. Deleting case 3 post-scoring yields 3/3 and2/3 but conceals failure if replacing original. Keep versions, raw outcomes and declared subgroup rules. No live model or broader quality measurement supplied.

- Six cases are planned; five run, four pass and one fails. One is skipped. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main manifest: six planned room-capacity cases C1–C6; case version classroom-v1, fixed lookup stub-v1. Inputs A/B/C/D/E/F have reference 12/8/10/6/4/2. Completed actual outputs 12/8/10/6/5 for C1–C5 yield four passes and one fail; C6 explicitly skipped with reason, not run. Completed pass rate 4/5=80%, coverage 5/6≈83.33%; skip remains unassessed.

  One explicit record per planned ID required; missing, duplicate, unknown IDs or unexpected fields reject the record set. Wrong actual output type/value is a completed failure. All explicit skips give coverage 0 and undefined completed pass rate. A stub replay rechecks the same procedure, not independent data.

  Separate picture correctness flags A[1, 1, 0, 1], B[1, 0, 0, 1] yield 75%/50% on four shared cases; case 2 differs, case 3 failsboth. Deleting case 3 post-scoring yields 3/3 and2/3 but conceals failure if replacing original. Keep versions, raw outcomes and declared subgroup rules. No live model or broader quality measurement supplied.

- A report omits this qualification: Missing evaluations can bias results; neither completion rate nor pass rate should hide the other.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main manifest: six planned room-capacity cases C1–C6; case version classroom-v1, fixed lookup stub-v1. Inputs A/B/C/D/E/F have reference 12/8/10/6/4/2. Completed actual outputs 12/8/10/6/5 for C1–C5 yield four passes and one fail; C6 explicitly skipped with reason, not run. Completed pass rate 4/5=80%, coverage 5/6≈83.33%; skip remains unassessed.

  One explicit record per planned ID required; missing, duplicate, unknown IDs or unexpected fields reject the record set. Wrong actual output type/value is a completed failure. All explicit skips give coverage 0 and undefined completed pass rate. A stub replay rechecks the same procedure, not independent data.

  Separate picture correctness flags A[1, 1, 0, 1], B[1, 0, 0, 1] yield 75%/50% on four shared cases; case 2 differs, case 3 failsboth. Deleting case 3 post-scoring yields 3/3 and2/3 but conceals failure if replacing original. Keep versions, raw outcomes and declared subgroup rules. No live model or broader quality measurement supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled evaluation pipelines **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of evaluation pipelines.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Evaluation pipelines. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Evaluation pipelines. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Eight planned cases, six complete, five pass, one fail. Give observed pass rate and coverage.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

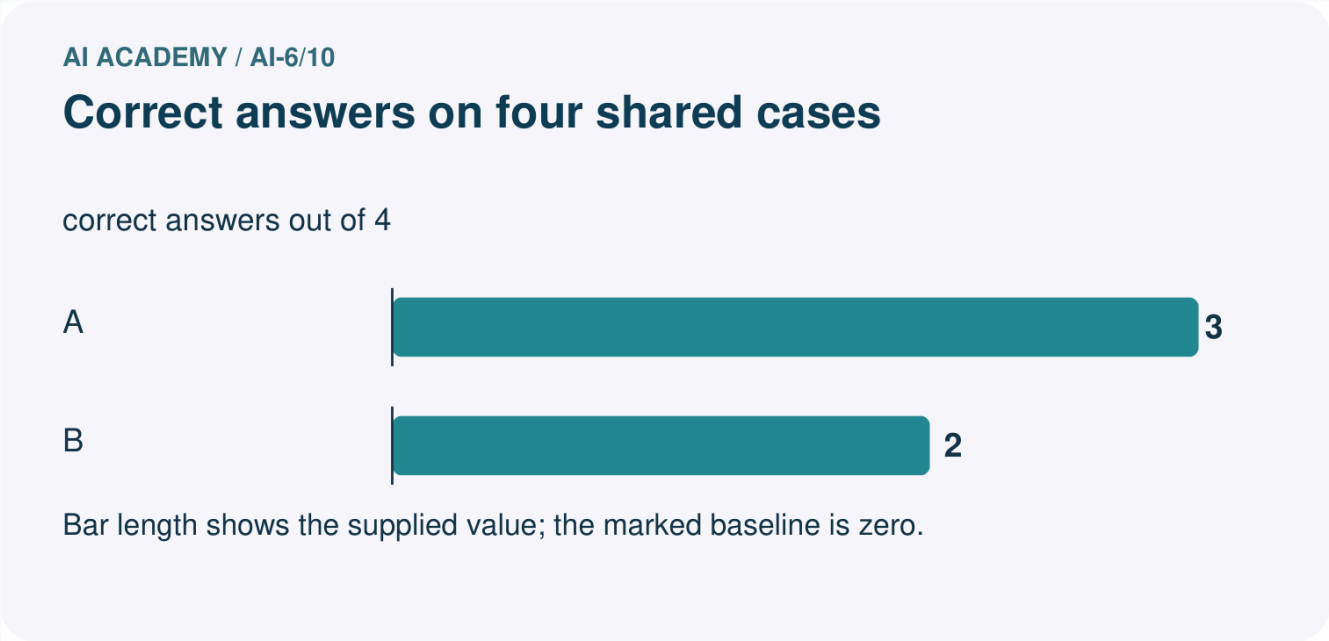
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.
- A scores [1,1,0,1]; B scores [1,0,0,1]. Each 1 means agreement with the reference answer.
- Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/10

Evaluation pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Freeze cases

Keep questions and reference labels unchanged for both versions.

2. Compute rates

A is 3/4=75%; B is 2/4=50%.

3. Inspect failures

Case two separates the versions; case three fails for both.

CHANGE ONE THING

Remove the shared failed third case after scoring.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels.
- Which labelled step matches this detail? Case two separates the versions; case three fails for both.
- 1. Freeze cases
- 2. Inspect failures
- 3. Compute rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect failures Case two separates the versions; case three fails for both. A pipeline preserves comparable cases, version labels and scoring.

Report the denominator and disagreements alongside percentages.

Changed-case question and optional hint

- Changed-case practice: Remove the shared failed third case after scoring. Which conclusion is supported by the supplied facts?
- 1. Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. The population changed after seeing failures, concealing a weakness.

Repair and retain the record

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

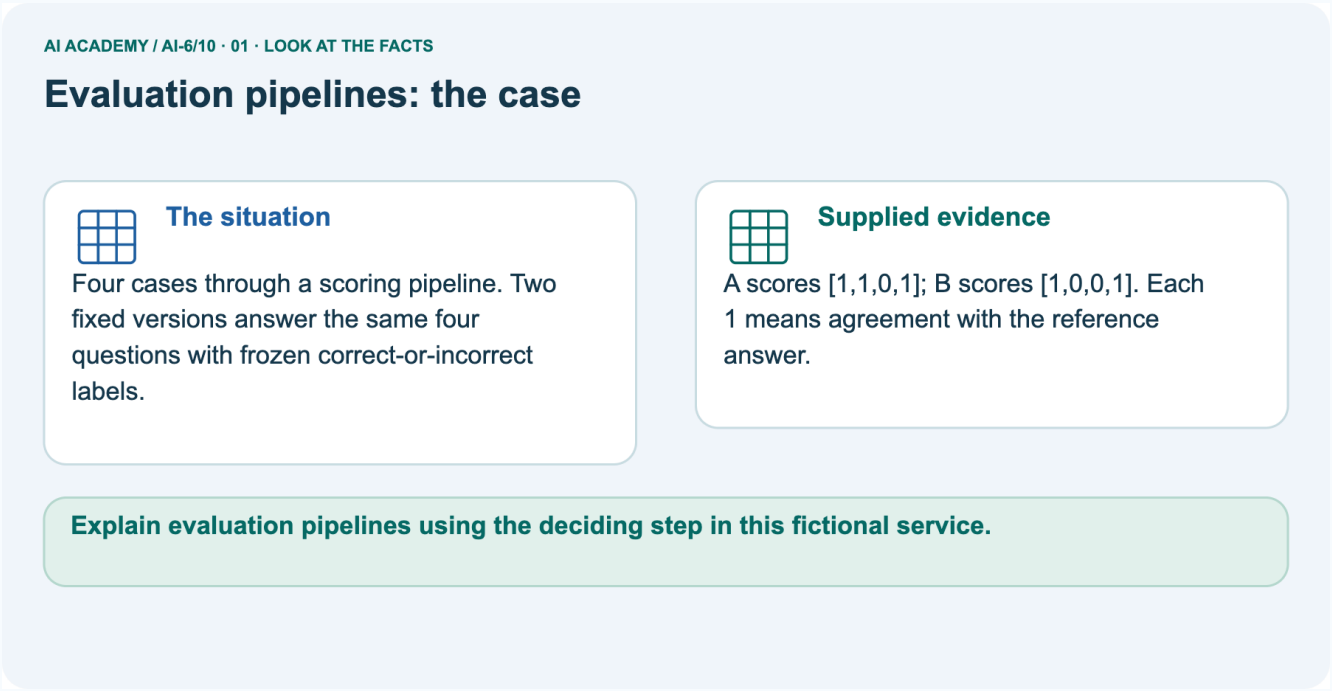
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Evaluation pipelines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

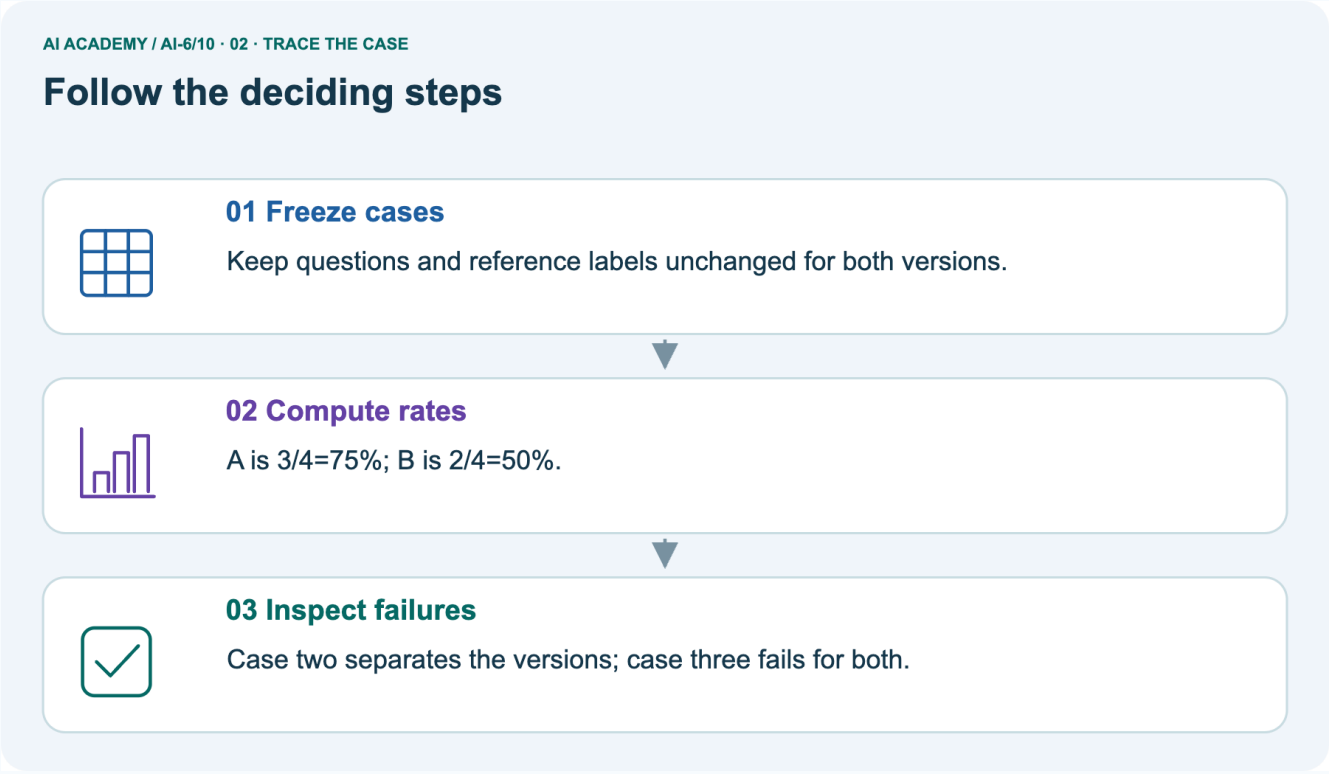
Four cases through a scoring pipeline. Two fixed versions answer the same four questions with frozen correct-or-incorrect labels. Supplied facts: A scores [1,1,0,1]; B scores [1,0,0,1].

Each 1 means agreement with the reference answer.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



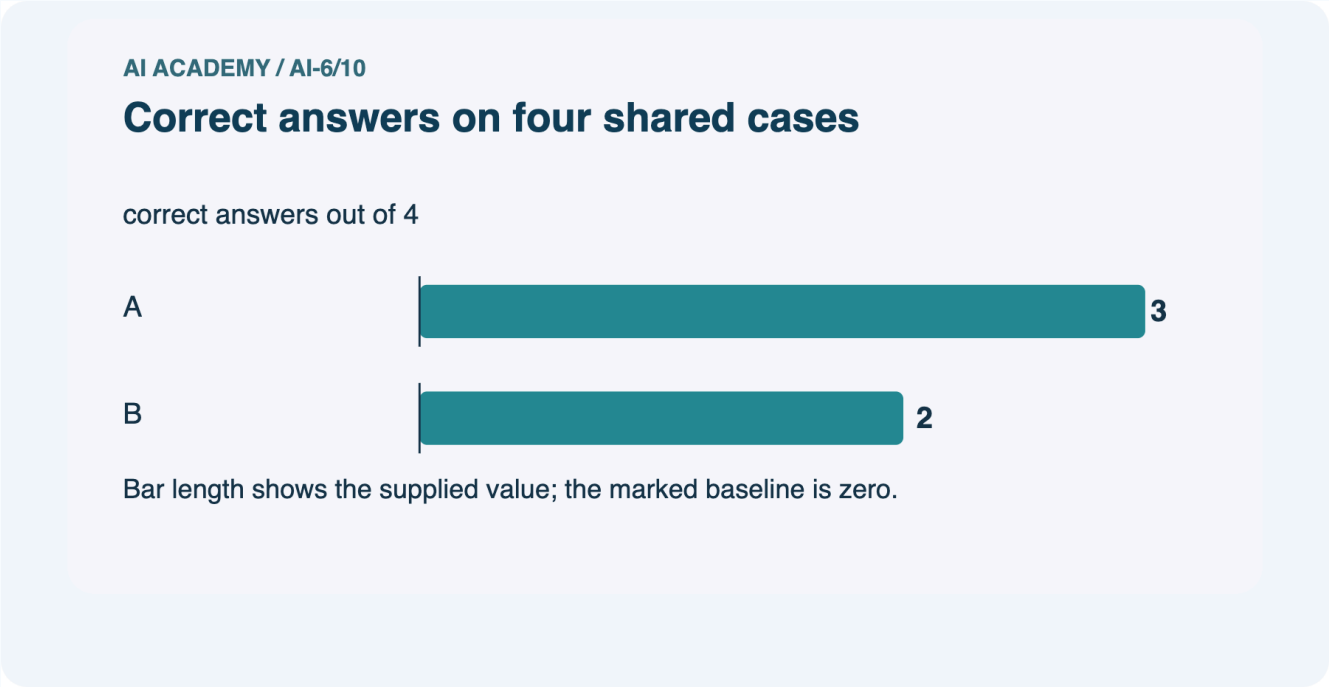
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze cases: Keep questions and reference labels unchanged for both versions. → Compute rates: A is 3/4=75%; B is 2/4=50%. → Inspect failures: Case two separates the versions; case three fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Correct answers on four shared cases



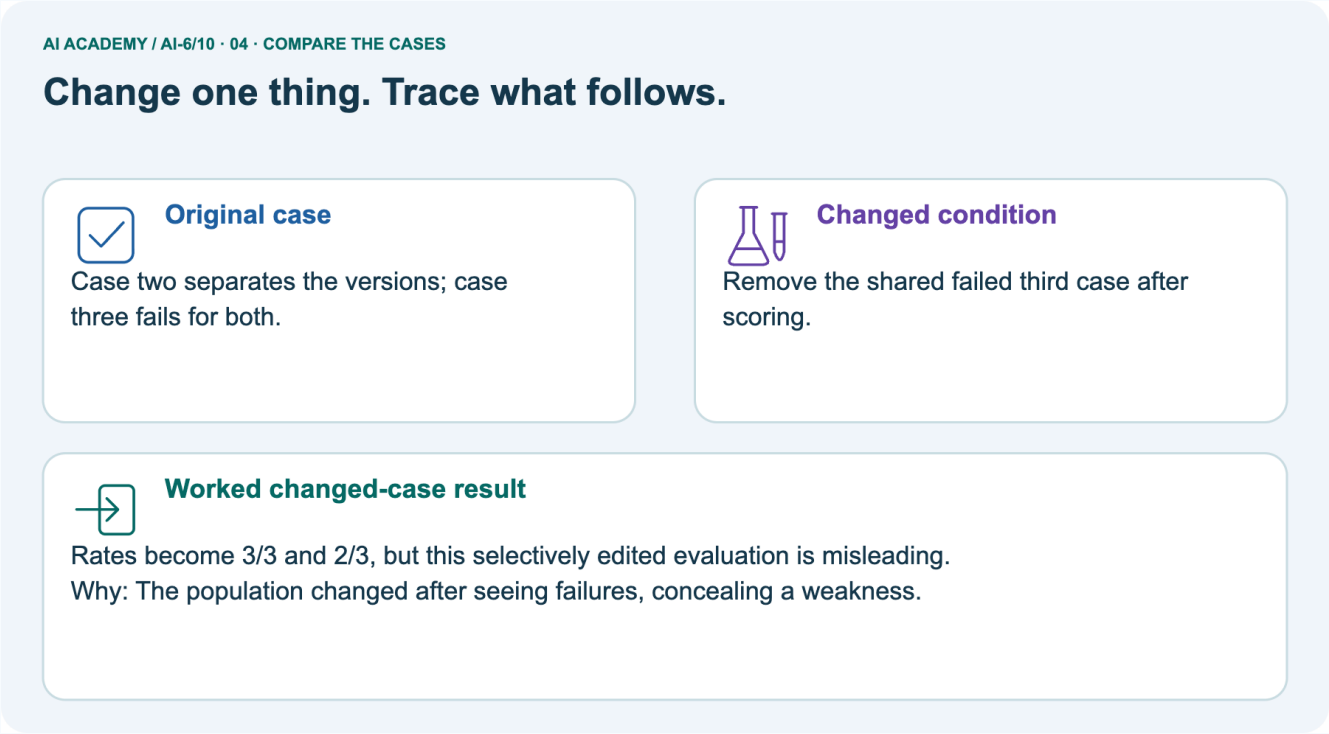
Follow the labelled picture with the matching explanation. Open the vector diagram

Correct answers on four shared cases. Units: correct answers out of 4. A: 3. B: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

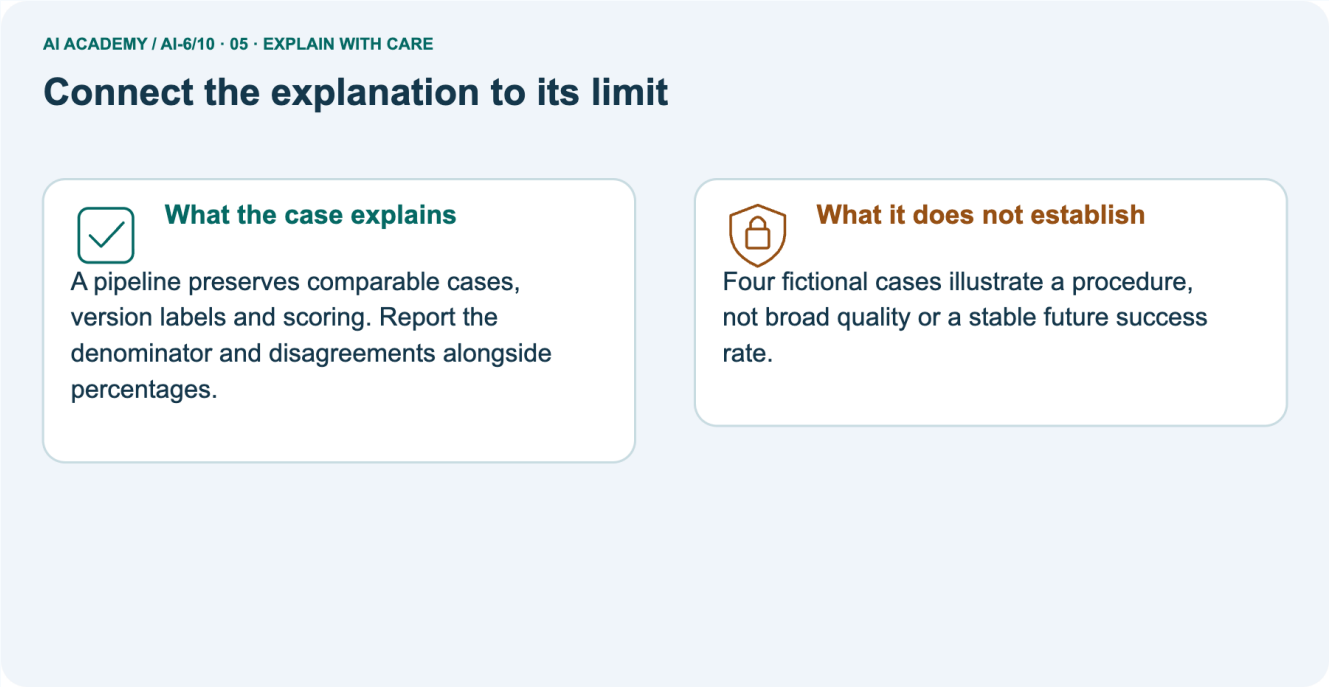
Original case: Case two separates the versions; case three fails for both. Changed condition: Remove the shared failed third case after scoring.

Expected result: Rates become 3/3 and 2/3, but this selectively edited evaluation is misleading. Reason: The population changed after seeing failures, concealing a weakness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

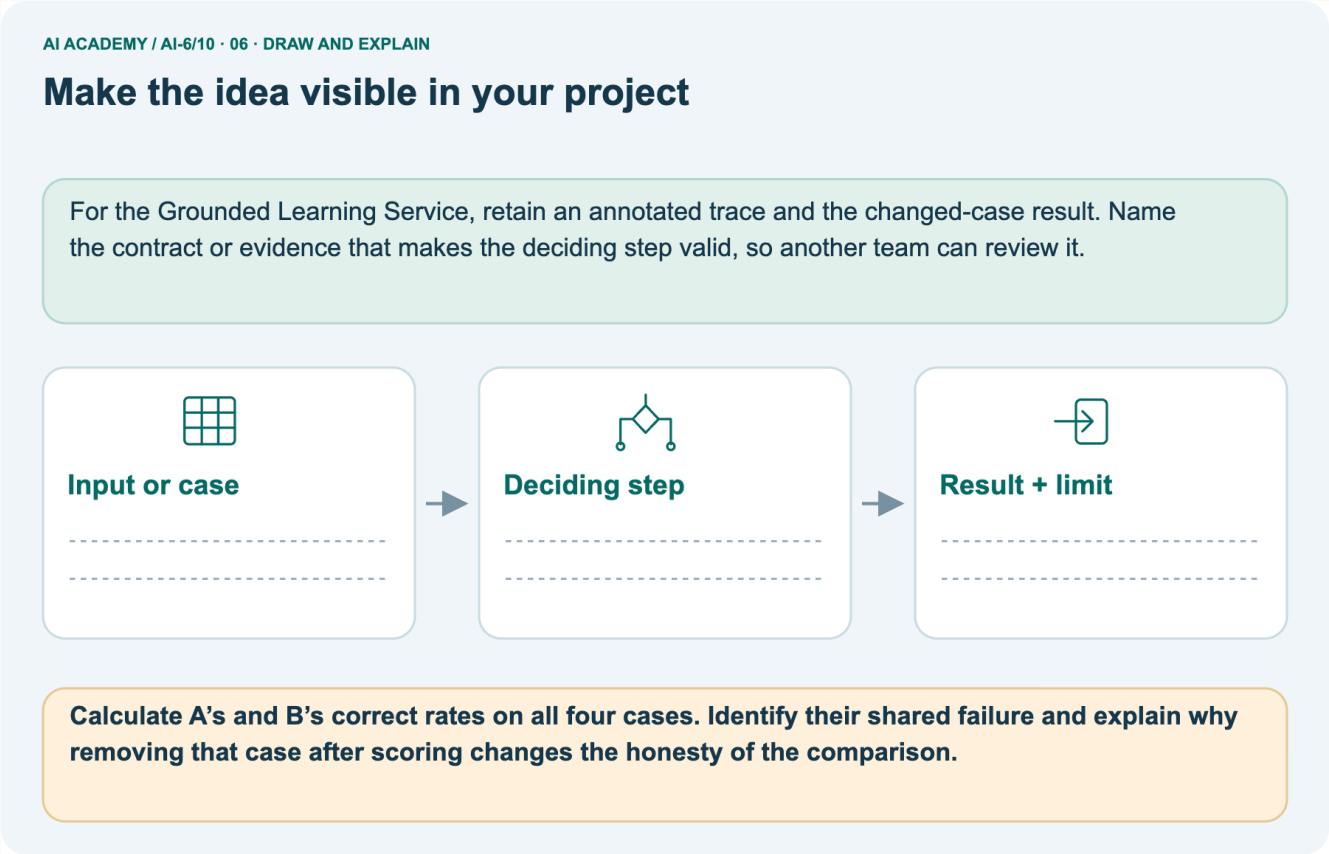
A pipeline preserves comparable cases, version labels and scoring. Report the denominator and disagreements alongside percentages.

Limit: Four fictional cases illustrate a procedure, not broad quality or a stable future success rate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Evaluation pipelines: the case

02. Follow the deciding steps

03. Correct answers on four shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.